

An Exact 7×7 Counterexample to the Lyapunov Norm Conjecture

Preliminary computational note

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This note records a computer-assisted but exact integer certificate. It does not make a claim of literature priority; independent verification and a full bibliographic search should precede any public novelty claim.

1 Statement

For $A \in \mathbb{R}^{n \times n}$, define the Lyapunov operator

$$\mathcal{L}_A(X) = AX + XA^T,$$

with domain and range equipped with the Frobenius norm. Let $\mathbb{S}^n$ and $\mathbb{K}^n$ denote the real symmetric and skew-symmetric subspaces. Both are invariant under $\mathcal{L}_A$ and are orthogonal, hence

$$\|\mathcal{L}_A\|_{\mathbb{F} \rightarrow \mathbb{F}} = \max\{\|\mathcal{L}_A|_{\mathbb{S}^n}\|_{\mathbb{F} \rightarrow \mathbb{F}}, \|\mathcal{L}_A|_{\mathbb{K}^n}\|_{\mathbb{F} \rightarrow \mathbb{F}}\}.$$

The Lyapunov norm conjecture asserts that the first restricted norm is always at least the second one.

Consider the integer matrix

$$A = \begin{bmatrix} 5 & 0 & -2 & 0 & 5 & 1 & 0 \\ 0 & 5 & 5 & -5 & -2 & -1 & -1 \\ 5 & 0 & 2 & 0 & 2 & 0 & -2 \\ 6 & 2 & 0 & 2 & -4 & -5 & 2 \\ -1 & -2 & 6 & 0 & 2 & 1 & -1 \\ 2 & 7 & -5 & -1 & 1 & 1 & -1 \\ -4 & 4 & 2 & 6 & 2 & -2 & -2 \end{bmatrix}.$$

We prove the strict inequalities

$$\|\mathcal{L}_A|_{\mathbb{S}^7}\|_{\mathbb{F} \rightarrow \mathbb{F}}^2 < 309 < \|\mathcal{L}_A|_{\mathbb{K}^7}\|_{\mathbb{F} \rightarrow \mathbb{F}}^2.$$

Consequently, this A is a counterexample to the conjecture.

2 Exact certificate on the symmetric subspace

Use the unnormalised ordered basis

$$\mathcal{B}_S = (E_{11}, \dots, E_{77}, E_{12} + E_{21}, E_{13} + E_{31}, \dots, E_{67} + E_{76})$$

of $\mathbb{S}^7$. Its Frobenius Gram matrix is

$$W = \text{diag}(I_7, 2I_{21}).$$

Let $R_S \in \mathbb{Z}^{28 \times 28}$ be the coordinate matrix of $\mathcal{L}_A|_{\mathbb{S}^7}$ in this basis. Thus, if s is the coordinate vector of $S \in \mathbb{S}^7$, then

$$\frac{\|\mathcal{L}_A(S)\|_F^2}{\|S\|_F^2} = \frac{s^T R_S^T W R_S s}{s^T W s}.$$

Define the integer symmetric matrix

$$M = 309W - R_S^T W R_S.$$

The 28 leading principal minors of M are listed in the appendix and are all strictly positive. Sylvester's criterion therefore gives $M \succ 0$. Hence, for every nonzero $S \in \mathbb{S}^7$,

$$\|\mathcal{L}_A(S)\|_F^2 < 309\|S\|_F^2,$$

which proves

$$\|\mathcal{L}_A|_{\mathbb{S}^7}\|_{F \rightarrow F}^2 < 309.$$

The accompanying Python verifier constructs R_S directly from A and computes all principal-minor determinants with arbitrary-precision integer arithmetic using fraction-free Bareiss elimination.

3 Exact skew-symmetric witness

Take

$$K = \begin{bmatrix} 0 & -16 & -1 & 2 & 1 & 2 & 1 \\ 16 & 0 & -3 & -2 & 1 & 1 & -1 \\ 1 & 3 & 0 & 9 & -1 & 0 & -1 \\ -2 & 2 & -9 & 0 & 1 & 0 & 2 \\ -1 & -1 & 1 & -1 & 0 & -5 & 1 \\ -2 & -1 & 0 & 0 & 5 & 0 & 0 \\ -1 & 1 & 1 & -2 & -1 & 0 & 0 \end{bmatrix} \in \mathbb{K}^7.$$

A direct integer calculation gives

$$\|K\|_F^2 = 794, \quad \|AK + KA^T\|_F^2 = 246134.$$

In particular,

$$246134 - 309 \cdot 794 = 788 > 0,$$

so

$$\frac{\|\mathcal{L}_A(K)\|_F^2}{\|K\|_F^2} = \frac{246134}{794} = \frac{123067}{397} > 309.$$

Therefore

$$\|\mathcal{L}_A|_{\mathbb{K}^7}\|_{F \rightarrow F}^2 > 309,$$

which, together with the symmetric certificate, disproves the conjecture.

4 Numerical values and discovery route

For orientation only, direct singular-value computations in orthonormal bases give

$$\|\mathcal{L}_A|_{\mathbb{S}^7}\|_{F \rightarrow F} \approx 17.563064546415227, \quad \|\mathcal{L}_A|_{\mathbb{K}^7}\|_{F \rightarrow F} \approx 17.632771343495410.$$

Their squared values are approximately 308.46123626154745 and 310.91462525199294, respectively. These floating-point values are not used in the proof.

The candidate was found by representing $\mathcal{L}_A$ separately on orthonormal bases of $\mathbb{S}^n$ and $\mathbb{K}^n$, optimising the logarithmic norm ratio, then rounding and locally searching over integer matrices. The final step replaced all floating-point evidence by the exact threshold certificate above.

A Leading principal minors of M

The basis ordering is the one stated in Section 2. Fraction-free Bareiss elimination gives the following exact leading principal minors $\Delta_j = \det(M_{1:j,1:j})$.

$\Delta_1 = 45,$	$\Delta_{15} = 4404454996722461231908772889088,$
$\Delta_2 = 2835,$	$\Delta_{16} = 551148090513833836614132476163584,$
$\Delta_3 = 272475,$	$\Delta_{17} = 82653964625151693689225801274556416,$
$\Delta_4 = 44318275,$	$\Delta_{18} = 14026385989709428992250275618351523840,$
$\Delta_5 = 7680534735,$	$\Delta_{19} = 1502392927970164695847925783204260347904,$
$\Delta_6 = 1814189012387,$	$\Delta_{20} = 157700921046263086513545457890190023065600,$
$\Delta_7 = 479566281444205,$	$\Delta_{21} = 14195367845885091720876040998833226175676416,$
$\Delta_8 = 19832528104131348,$	$\Delta_{22} = 2213057904866526824695626989288024042102849536,$
$\Delta_9 = 1540829913299718500,$	$\Delta_{23} = 277858161842696190209887882976208033316887330816,$
$\Delta_{10} = 157363938670919564408,$	$\Delta_{24} = 17196521011372249832152125202172888600255681003520,$
$\Delta_{11} = 18863824393258624503728,$	$\Delta_{25} = 3431647585288386392445666119906081353182870635544576,$
$\Delta_{12} = 3346138427793119078031776,$	$\Delta_{26} = 468342744962523850627272604389662272814886119641448448,$
$\Delta_{13} = 735531340703540071925600768,$	$\Delta_{27} = 26729243136733672938386009039494448624212842052608589824,$
$\Delta_{14} = 62410852596115274619525697920,$	$\Delta_{28} = 5086805439078841307614940410877708960083347249811572129792.$

Every Δ_j is positive, completing the exact Sylvester certificate.